Pharmaceutics

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che-50

2022, 2020, tu

⊛ What is the aim of solubility study? Discuss the conditions used for different type solubility?

⇒ The aim of the solubility study is to ensure that the released products are maintained within the specification limit through their entire shelf life.

⇒ There are two types of stability study: Real time stability study & accelerated stability study.

According to the international conference on harmonization:

Study	Condition	Duration	sequence of testing
Real time	Temp. 25±2°C RH 60±5%	up to self life	3 month duration in 1st year 6 month duration in 2nd year once in third year
Accelerated	Temp. 40±2°C RH 75±5%	6 month	Done initially Done in 3 month Done in 6 month

2021

⊛ Briefly discuss the steps involved in wet granulation?

⇒ wet granulation form granules by binding the

Powder together with an adhesive instead of Compaction

Wet granulation consist several steps.

There are five steps involves in wet granulation:

Step-1: Mixing of the drug and excipient in powder fo

Step-2: Mixing of the binder solution in powder mix to form wet mass. there is a limit to the amount o the solvent that may be employed.

Step-3: The wet mass convert into coarse granular aggregates by passing through a hammer mill, equipped with screen having large perforation.

Step-4: A drying process is required in all wet granulatio procedures to remove the solvent that was used in formi the aggregates & to reduce the moisture content to an optimum level of concentration level within the granules.

Step-5: After drying screening of the granules again. The size of the screen depends upon the

grinding equipment used & size of the tablet to be made.

2020, 2022, tu

(*) what are the commonly used pharmaceuticals doges forms? State. Advantage of tablet over other ~~d~~dages?

⇒ The most commonly used pharmaceutical doges form are: 1. Solid: tablet, capsule, powder.

2. Semi solid: cream; paste, Gel, Suppositories.

3. Liquid: Monophasic: syrup, solution

Biphasic: Emulsion, Suspension.

4. Gas: Inhaler, Aerosols.

⇒ Tablet may be defined as the solid unit dosage form of medication with suitable excipients. It comprises a mixture of active substances and excipients, usually in powder form, that are pressed or compacted into a solid dose. Advantage of tablets:

① They are unit dosage form and offer the greatest capabilities of oral dosage forms with the highest dose prescriptions & the least content variability.

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2. Their cost is lowest of all oral dosage forms.
3. They are in general, easiest & cheapest to packaging & shipment.
4. product identification is simplest & cheapest.
5. They lend themselves to certain special release profile such as enteric & delayed release product
6. They are better suited to large scale production than other unit dosage form.
7. they have the best combined properties of chemi mechanical & microbiological stability of all the oral form.

(★) 2020, 2016
what is coating? Briefly discuss types of coating?
or, short note of coating?

⇒ Coating is an additional steps in pharmaceutical manufacturing which means masking the taste of drug. coating is applied in many purposes such as: t mask the taste, odor of the drugs, to provide physical & chemical protection of drug ~~etc~~, to control the release of the drug from the tablet.

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⇒ Coating is classified into three major catagories:

1. Sugar coating: This involves the application of sugar based coating solution to the tablets. this coating is water soluble after swallowing. The sugar coating involves the following steps: (i) Sealing (ii) Sub coating (iii) Syruping (iv) Finishing (v) polishing.

2. Film coating: In this process there is a deposition of thin film of polymer surrounding the tablet core. The film formed is very thin & it gives the tablet a less weight & small size. EX. EC(Ethyl cellulose), HPC(Hydroxy propyl cellulose), HPMC(hydroxy propyl methyl cellulose) etc, are use in film coating. Film coating divided into two classes: (1) Aqueous coating (2) Organic coating.

3. Enteric coating: enteric coating tablets have delayed release features. They are designed to pass unchanged through the stomach to the intestines, where the tablets disintegrates & allow drug dissolution and absorption. Example: omeprazole, Aspirin etc.

⇒ Coating process: Tablet coating is the application of

a coating composition to a moving bed tablet with concurrent use of heated air to facilitate the evaporation of the solvent. The distribution of the coating is accomplished by the movement of the tablets perpendi (coating pan) or vertical (air suspension coater) to the application of the coating composition. Most coating proces use one of three general types of equipments: 1) th Standard coating pan, 2) the perforated coating pan, 3) the fluidized bed (air suspension).

The source of defect and other probable problems of film defect of coating: Sticking and picking, Roughness, orange peel effect, Bridging and filling, Blistering, Hazing, color variation, cracking.

⊛ 2022 predict effect of following parameter on properties of tablets: (i) particle size & shape (ii) surface area (iii) density

⇒ (i) particle size & shape: particle size of granulation can affect the tablet weight, tablet weight variation, disintegration time, granules friability, flow ability

and drying rate kinetics of wet granulation.

(ii) Surface area: Drugs have very limited water solubility. So the finely milled powdered increase surface area can affect the rate of dissolution.

(iii) Density: Granules density may influence compressibility, tablet porosity, dissolution and other properties. Dense and hard granules may require high compressible loads to produce cohesive compact which increase the disintegration and dissolution time.

tutorial

(X) Write the purpose of coating?

⇒ purpose of coating:

1. To mask the taste, odor of the drugs.
2. To provide the physical and chemical protection of drugs.
3. To control the release of the drug from the tablet.
4. To protect the drug from the gastric environment of the stomach with an acid resistance enteric coating.
5. To improve the pharmaceutical elegance by use of special color or coating printing.

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Tu, 2022

(★) What is coating? write the causes and solution of the following coating problems: (i) orange-peel effect, (ii) Blistering, (iii) Cracking?

⇒ (i) orange-peel effect:

Reason: Inadequate Spreading of coating solution before dry cause orange-peel effect on the coating. This is due to the rapid drying or by high solution viscosity.

Solution: Thinning the solution with additional solvent may Solve this problem.

(ii) Blistering:

Reason: when coated tablets requires further drying in oven, too rapid evaporation of the solvent from the coat and the effect of high temperature on strength, elasticity and adhesion of the film may result in Blistering.

Solution: Milder drying condition are warranted in this case.

(iii) Cracking:

Reason: cracking occur if internal stressed is the film exceed the tensile strength of the film. The tensile strength of the film can be increased by using higher molecular weight polymer blends.

Solution: Internal stress in the film can be minimized by adjusting the plasticizer type, concentration, pigment type and concentration.

⊛ 2018 what are the problems in tablet coating?

⇒ Problems are: Sticking and picking, Roughness, orange peel effect, Bridging and filling, Blistering, Hazing, Color variation, Cracking.

⊛ 2015, tu write the disadvantage of tablets?

⇒ 1. Some tablets may have unpleasant taste and odor, which can make them difficult for patient to swallow.

2. Tablets can be large and difficult to swallow, especially for children, elderly patients, or those with swallowing disorder.

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3. Some tablets can irritate the gastrointestinal tract.

4. Some active ingredients may not be suitable for tablet formulation due to chemical instability.

5. Tablets can breakdown due to factors like temperature fluctuations, mechanical stress and exposure to light. This can affect the dissolution rate and drug release profile.

6. Some drug resist compression into tablet form due to their amorphous nature.

7. Drugs with poor wetting, slow dissolution properties, may be difficult to formulate a tablet.

2018, 2022

⊗ Define capsule? Give its advantage & disadvantage?

⇒ Capsules are a solid dosages form in which drug substance is enclosed in a small gelatinous shell. The capsules are available both as hard capsule & soft capsule.

⇒ Advantage:

1. Have elegance, ease of use & portability.
2. They are produced economically in large quantities wide range of colors.

3. capsule provide a smooth, slippery, easily swallowed & tasteless shell for drugs.

4. They are attractive in appearance.

5. They are easy to handle and carry.

6. packaged & shipped at low cost with less breakage.

7. capsules are tasteless, odorless & can easily be administered.

Disadvantage:

1. Difficult to swallow for children & older people.

2. capsules which made of gelatin, can be sensitive to environmental factors like humidity & temperature, which can affect their integrity & shelf life.

3. Manufacturing cost is more than tablets.

4. capsule have a limited capacity, which can restrict the dosage amount.

5. Manufacturing process can be more complex than tablet.

6. capsule should not be used for highly efflorescent materials.

2021

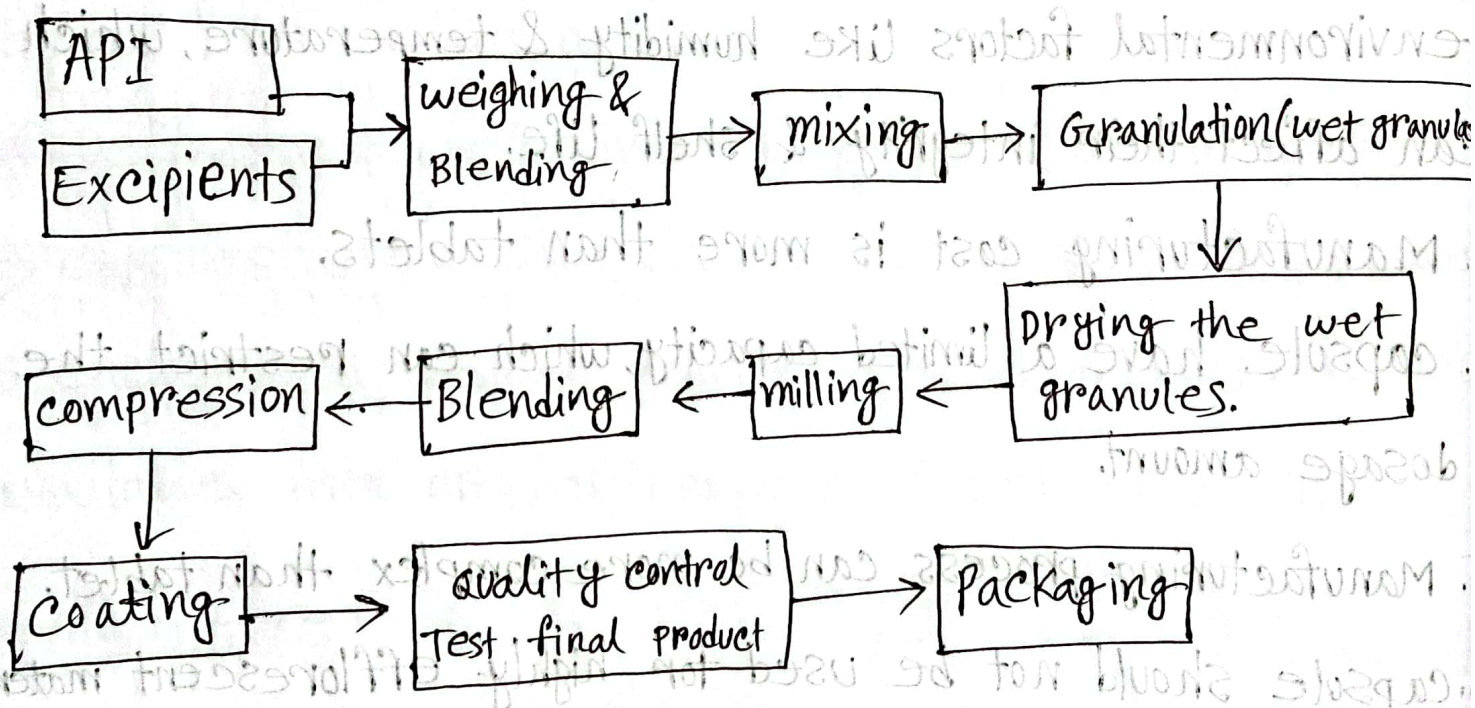
What is 'referance standard?

⇒A drug reference standard or pharmaceutical reference standard is a highly characterized material suitable

to test the identity, strength, quality & purity of substances for pharmacetical use and medicinal products.

2018, 2019(t)
What is granulation? Give the flow chart of unit proces in wet granulation method?

⇒ Granulation is a part of pharmaceutical process that attempts to improve the flow of powdered materials by forming sphere like or regularly shaped aggregates called granules. Flow chart of wet granulation:



TU
What is the API of Vit-C & Aspirin?

⇒ API of Vit-C is L-Ascorbic Acid. And API of Aspirin is acetylsalicylic acid.

2017, 2015

Compare wet & dry granulation methods for tablet manufacture

Wet	Dry
It is the process of forming granules by adding a granulating liquid.	It is the process of forming granules without using any liquid solution.
Requires powder particles, a granulating fluid & a sieve.	Requires finely powdered compound & tablet pressers or roll compactors.
Involves mixing the powder with a granulating fluid, followed by forcing through a sieve to make tablets.	Involves the direct compression of a finely ground powder.
use of solvent	No use of solvent.
use of wet binder	use of dry binder.
Heat & Temperature are required for drying of granules	No heat & Temperature are required for drying of granules.
requires more equipment, more space, & higher investment cost	requires less equipment, less space & less cost.
It is difficult to scale up.	It is reproducible & scalable.
Generally provides uniform granules	Can produce variable granule size but milling can help improve uniformity.

(X) 2017 point out the difference btwn tablet & capsule?

Tablet	capsule
Solid, compressed powder.	Gelatin or HPMC shell containing powder, liquid, or pellets.
Various shapes (round, oval, oblong)	Typically cylindrical or oval.
Can be larger & more varied in size	Generally smaller & easier to swallo
High precise dosages.	High but some variability with powder settling.
Tablet Dissolves gradually in the stomach or intestines.	Gelatin/HPMC shell dissolves quickly, releasing contents
Generally more stable, less sensitive to environmental conditions	Shells can be sensitive to himidity & temperature.
Usually longer shelf life.	May have shorter shelf life due to shell sensitivity.
Difficult to mast taste, coating or flavors needed.	Shell masks taste effectively.
Generally lower production cost	higher P. cost due to shell material
Limited to stable powders and granules.	Can contain powders, liquids, pellets, or semi-solids.
can be harder to swallow, especially larget tablets	easier to swallow, than tablet.

2017, 2019

(x) Give the reason for using gelatin for the formulation of capsule shell?

⇒ Reason: (1) It is highly biocompatible & digestible, ensuring safe & effective delivery of the drug.

(2) Gelatin forms a flexible & elastic ~~cett~~ shell that can easily be filled with various drug formulations, including powders, granules and liquids.

(3) Gelatin is a relatively inexpensive material, making it a cost-effective choice for capsule production.

(4) It dissolves readily in the stomach, allowing for the timely release of the API.

(5) It has excelent film-forming properties, allowing it to create a solid & stable shell.

(6) Gelatin can be easily molded into various shapes & size, making it suitable for different types of capsule.

(7) It generally has a neutral taste & odor, which helps in masking the taste of the encapsulated contents.

(8) Gelatin capsules can be used for both solid & liquid formulations, providing versatility in drug delivery options.

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⭐ 2015 What is the purpose of granulation in tablet manufacture

⇒ Purpose: ① Improved flow properties: Granules have better flow properties compared to fine powders.

② Enhanced compressibility: Granules are more compressible than powder.

③ Uniformity: Granulation helps to increase the uniformity of the API throughout the dosage form.

④ Dust control ⑤ Improved solubility & dissolution.

⑥ prevention of segregation: Granulation minimizes segregation by forming uniform granules with consistent particle size and density.

⑦ Enhance stability: Granulation helps to improve the stability of the tablet formulation.

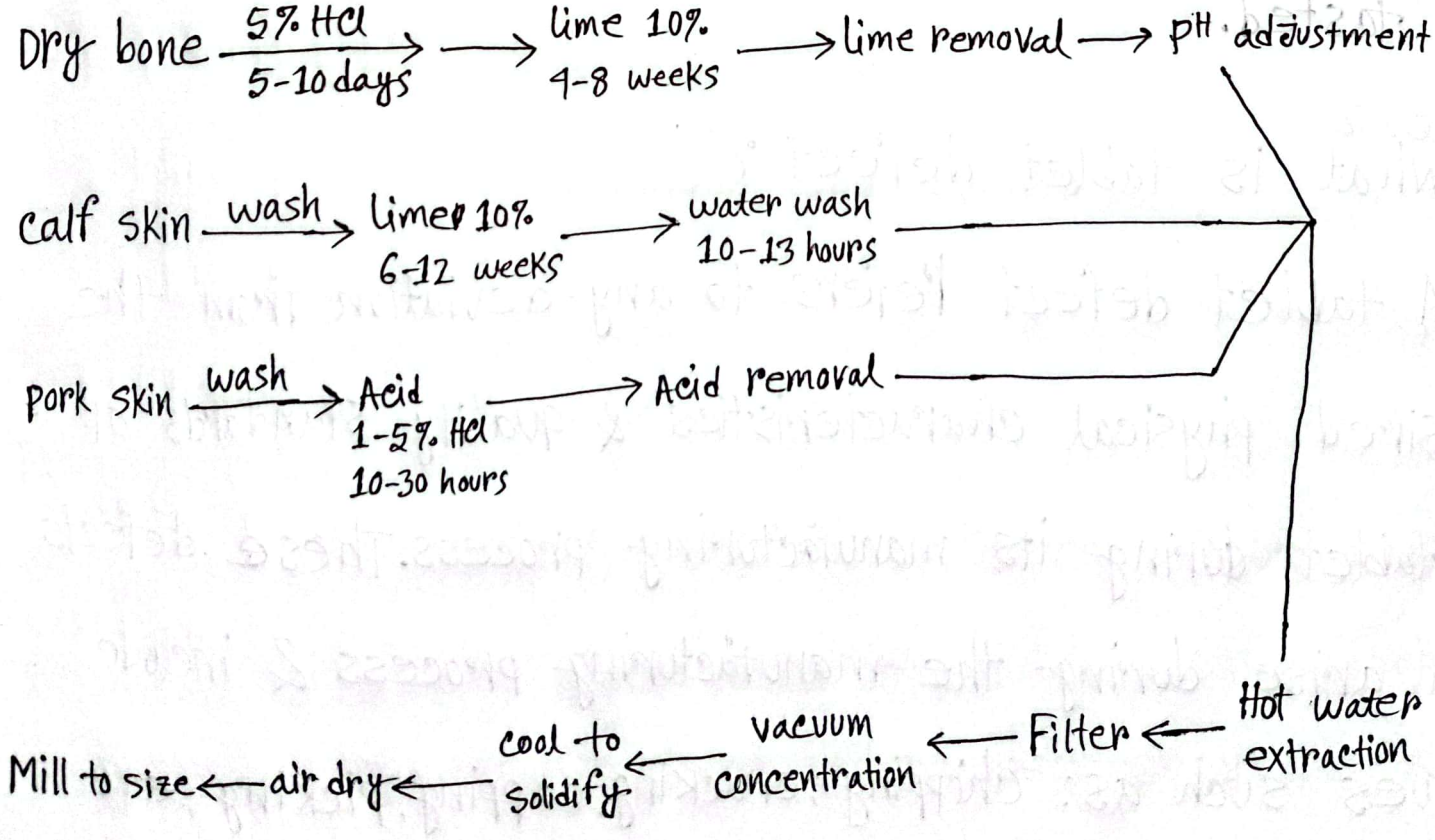
⭐ 2022, Define soft gelatin capsule (2015, Tu, 2019(1)) ? what is hard gelatin? outlin the synthesis (2018) of gelatin ?

⇒ A soft gelatin capsule is a type of capsule used to contain liquid or semisolid substances. The outer shell is made of gelatin, water & a plasticizer like glycerin

or sorbital, making it flexible & easily to swallow. It may contain additional ingredient such as preservative, coloring & opaquing agents, flavoring agents, sugar, acid & medicament to achieve desired effects.

⇒ A hard gelatin capsule is a type of capsule that is used to contain medicine in the form of dry powder or very small pellets. Hard gelatin capsules are usually filled with powders, granules, or tiny pellets. These capsules consist of two parts a body & a cap, both made from gelatin, water & sometimes other ingredients like dyes or preservatives.

⇒ A flow diagram of hard gelatin synthesis:



2015
⊛ what is ophthalmic preparations? why do we taste our eye drops?

⇒ ophthalmic preparations are sterile products that are intended to be applied to the eyelids or placed in the Space btwn eyelids & eye ball. Many ophthalmic preparatio are used in the practice of medicine.

⇒ It is due to the drainage of the solution through the nasolacrimal duct, which connects the eyes to the nasal cavity. From the nasal cavity, the liquid can travel down the back of the throat to the mouth, where you can taste it. The eye drop travel by the path so sometimes be tasted.

2022
⊛ what is tablet defect?

⇒ A tablet defect refers to any deviation from the desired physical characteristics & quality standards of a tablet during its manufacturing process. These defects can arise during the manufacturing process & include issues such as: chipping, cracking, capping, picking etc.

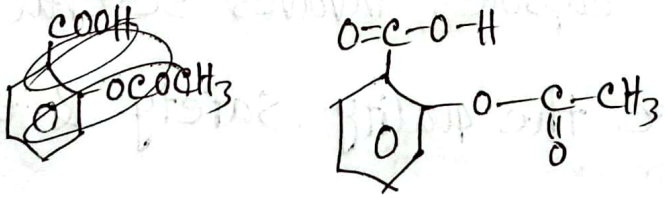
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2021, 2017
⊗ write the necessary condition for ophthalmic preparation?

⇒ several necessary conditions must be meet to ensure their safety, effectiveness, & patient comfort. These conditions include: Sterility, Isotonicity, Proper pH, clarity viscosity, preservative content, stability, Non-Irritating. Meeting all these necessary conditions ensures the safety & efficacy of ophthalmic preparations, protecting the delicate eye tissues & promoting optimal therapeutic outcomes.

2018
⊗ Mention API of Aspirin & draw its structure?

⇒ API of aspirin is acetylsalicylic acid. Structure:



acetylsalicylic acid (aspirin).

2018, 2019 (1)
⊗ What is major component of a capsule shell & why it is chosen?

⇒ Major component is gelatin. It is chosen because of several reason: Biocompatibility, solubility, Film-forming properties, Flexibility & strength, transparency, Inertness,

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Cost-effectiveness etc. Gelatin provides a safe, effective & convenient method for delivering drugs.

(X) IV define pharmaceutics? write evaluation of capsule?

⇒ pharmaceutics is the discipline of pharmacy that deals with the process of turning a new chemical entity (NCE) or old drugs into a medication to be used safely & effectively by patients. pharmaceutics helps relate the formulation of drugs to their delivery & disposition in the body.

⇒ The evaluation of capsules involves several tests & assessments to ensure the quality, safety & efficacy of the final product. The key evaluation parameters include: weight variation, content uniformity, Disintegration test, Dissoluti test, moisture content, hardness, Microbial Limits, stability testing, odor & taste, capsule integrity etc. Ensuring that capsules meet all necessary criteria is crucial for maintaining product quality, patient safety & therapeu

efficacy.

⭐ TU Write about soft gelatine capsule manufacturing process?

Medicament preparation
Gelatin Mass preparation → Encapsulation process
↓
capsule pre-shaping, washing and semi-drying
↓
capsule Drying
↓
capsule Inspection
↓
capsule Printing
↓
Quality assurance → packing → storage → press condition.

⭐ 2016 Write about requirement & formulation of ophthalmic preparation? - Short note?

⇒ O. preparation are sterile products that are intended to be applied to the eyelids or placed in the space btwn eyelids & eye ball.

Requirements of O.P.: Sterility, Isotonicity, proper pH, clarity, viscosity, preservative content, Stability, Non-Irritating, Packaging etc.

preformulation of ophthalmic preparations: the formulation of

ophthalmic preparations involves selecting appropriate excipients and processes to meet the above requirements. Different types of ophthalmic preparations include solutions, suspentions, ointments, gels, & inserts.

Solutions are the most common form of ophthalmic preparations. ophthalmic suspentions used when the active ingredient is not soluble in water. ophthalmic ointments provide prolonged contact time with the ocular surface. o. gels are viscous, providing sustained release & longer contact with the eye. o. inserts are solid or semi-solid devices that release the drug over time when placed in the eye.